

CS/PHYS 314/300: Quantum Computing: Homework #1

Dr. Faisal Alvi

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Problem 1

(10 points) Given a qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, and an operator denoted by a matrix U , prove that

- (a) (4 points) if U is unitary, then U preserves the norm of the qubit after application, i.e., $\| |\psi\rangle \| = \| U |\psi\rangle \|$.
- (b) (3 points) the rows of U are orthonormal to each other.
- (c) (3 points) the columns of U are orthonormal to each other.

Solution

- (a) $\| |\psi\rangle \| = \| U |\psi\rangle \|$ we take the square norm of the quantum state after applying the Unitary operation

$$\| U |\psi\rangle \|^2 = (U |\psi\rangle)^\dagger (U |\psi\rangle)$$

$$(U |\psi\rangle)^\dagger = \langle \psi | U^\dagger$$

$$\text{Therefore, } \langle \psi | U^\dagger U | \psi \rangle \stackrel{U^\dagger U = I}{=} \langle \psi | I | \psi \rangle = \langle \psi | \psi \rangle = \| |\psi\rangle \|^2$$

$$\| U |\psi\rangle \|^2 = \| |\psi\rangle \|^2$$

$$\| U |\psi\rangle \| = \| |\psi\rangle \|$$

Thus we have proven that unitary operations on a qubit do not affect its total probability, i.e. sum of the new probabilities will always equal 1.

- (b) rows are orthonormal if:

Norm of the row is 1.

Inner Product between rows is 0.

$$\text{For a 2x2 matrix, } UU^\dagger = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Such that, $\|r1\| = \|r2\| = 1$, and $\langle r1, r2 \rangle = 0$. Where $\|r1\|^2 = |a|^2 + |b|^2$, $\|r2\|^2 = |c|^2 + |d|^2$, and $\langle r1, r2 \rangle$ is the inner product between $r1, r2$: $ac^* + bd^*$, $ca^* + db^*$.

If the resulting matrix of any such arbitrary matrix U matches the Identity Matrix, we have proven that U 's rows are orthonormal to each other as $UU^\dagger = I$, this logic can be extended to n dimensions. .

- (c) Columns are orthonormal if:

Norm of the columns is 1.

Inner Product between columns is 0.

$$\text{For a 2x2 matrix, } U^\dagger U = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Such that, $\|c1\| = \|c2\| = 1$, and $\langle c1, c2 \rangle = 0$. Where $\|c1\|^2 = |a|^2 + |c|^2$, $\|c2\|^2 = |b|^2 + |d|^2$, and $\langle c1, c2 \rangle$ is the inner product between $c1, c2$: $a^*b + c^*d$, $b^*a + d^*c$. If the resulting matrix of any such arbitrary matrix U matches the identity Matrix, we have proven that U 's columns are orthonormal to each other as $U^\dagger U = I$, this logic can be extended to n dimensions.

Problem 2

(10 points) Consider a Bloch Sphere representation of a qubit $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$ as shown here.

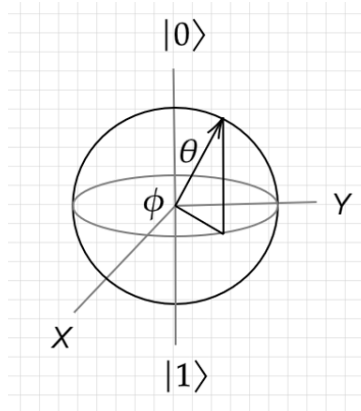


Figure 1: Bloch sphere representation of an arbitrary qubit state $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$

On the Bloch Sphere, plot the following qubits (show a plot similar to a diagram above):

(a) (2 points)

$$\frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle$$

(b) (3 points)

$$\frac{1-i}{2}|0\rangle + \frac{1+i}{2}|1\rangle$$

Next, consider the following operations:

$$(i) R(\theta_1) = \begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) \end{bmatrix}, (ii) S(\phi_2) = \begin{bmatrix} e^{-i\phi_2} & 0 \\ 0 & e^{i\phi_2} \end{bmatrix}$$

Given an arbitrary qubit on the X - Z plane (i.e.) its amplitudes have no imaginary component,

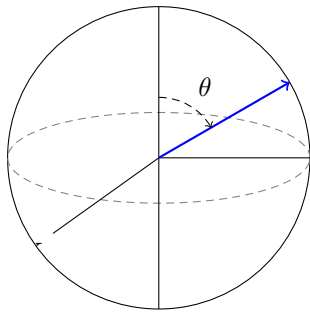
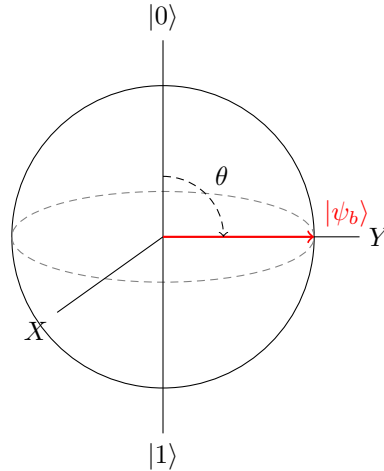
(c) (2 points) what is the effect of applying $R(\theta_1)$ to it? In particular state the effect when $\theta_1 = \pi/2$?

(d) (3 points) what is the effect of applying $S(\phi_2)$ to it? In particular state the effect when $\phi_2 = \pi/2$?

Solution

(a) $\frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle$
 $\cos(\frac{\theta}{2})|0\rangle + e^{-i\frac{\pi}{2}}\frac{1}{2}\sin(\frac{\theta}{2})|1\rangle$
 $\theta = 2\cos^{-1}(\frac{\sqrt{3}}{2}) = 60 \text{ degrees}, 1.047\text{rad}, \phi = \frac{\pi}{2} = 90 \text{ degrees}$

(b) $\frac{1-i}{2}|0\rangle + \frac{1+i}{2}|1\rangle$
 $e^{-i\frac{\pi}{4}}\left(\frac{1}{\sqrt{2}}|0\rangle + e^{i\frac{\pi}{2}}\frac{1}{\sqrt{2}}|1\rangle\right)$
 $\cos(\frac{\theta}{2})|0\rangle + e^{i\frac{\pi}{2}}\sin(\frac{\theta}{2})|1\rangle$
 $\theta = 2\cos^{-1}(\frac{1}{\sqrt{2}}) = 90 \text{ degrees}, 1.571\text{rad}, \phi = \frac{\pi}{2} = 90 \text{ degrees}$

(a): $\theta = \pi/3, \phi = \pi/2$ (b): $\theta = \pi/2, \phi = \pi/2$

- (c) The arbitrary qubit on the X-Z plane can be defined as: $|\phi\rangle = \alpha|0\rangle + \beta|1\rangle$ $\alpha, \beta \in \mathbb{R}$.
Therefore the qubit doesn't consist of any imaginary component.

$$R(\theta_1)|\psi\rangle = \begin{pmatrix} \cos(\theta_1) & -\sin(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha \cos(\theta_1) - \beta \sin(\theta_1) \\ \alpha \sin(\theta_1) + \beta \cos(\theta_1) \end{pmatrix}$$

As the all componenets are real, the resulting transforamtion rotates the qubit within the X-Z plane,
When $\theta_1 = \pi/2$, the qubit is rotated by 90 degrees around the y-axis (in the X-Z plane.)
the 90 degrees rotation transforms the qubit to $|\psi\rangle = -\beta|0\rangle + \alpha|1\rangle$.

- (d) The arbitrary qubit as defined in the question above, when applied.

$$S(\phi_2)|\psi\rangle = \begin{pmatrix} e^{-i\phi_2} & 0 \\ 0 & e^{i\phi_2} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha e^{-i\phi_2} \\ \beta e^{i\phi_2} \end{pmatrix}$$

Factoring out a global phase of $e^{-i\phi_2}$, the state becomes $\alpha|0\rangle + \beta e^{i(2\phi_2)}|1\rangle$. This operation rotates the qubit around the Z-axis by an angle of $2\phi_2$ introducing a complex phase in the Y-axis.

Problem 3

(10 points) So far we have studied qubits defined using two basis states: $|0\rangle$ and $|1\rangle$. We have also seen two related states: the $|+\rangle$ state and the $|-\rangle$ state. Answer the following questions with reasons:

- (a) (2 points) Do the $|+\rangle$ state and the $|-\rangle$ state form a pair of orthonormal states?
- (b) (2 points) Can the $|+\rangle$ state and the $|-\rangle$ state be used as the basis states?
- (c) If the answer to the both (a) and (b) is Yes, express the following qubits as a linear combination of the $|+\rangle$ state and the $|-\rangle$ state:
 - (i) (2 points) $|0\rangle$,
 - (ii) (2 points) $|1\rangle$,
 - (iii) (2 points) an arbitrary qubit: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

Solution

- (a) for $|0\rangle, |1\rangle$ to be a valid pair of orthonormal states, we need them to be orthogonal to each other (i.e. their inner product equals 0) and normal (they both must have unit norm).

Representation:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

Orthogonality:

$$\langle + | - \rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{2}} \times -\frac{1}{\sqrt{2}} \right) = \frac{1}{2} + \frac{-1}{2} = \frac{1}{2} - \frac{1}{2} = 0$$

$$\langle - | + \rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \right) + \left(-\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \right) = \frac{1}{2} + \frac{-1}{2} = \frac{1}{2} - \frac{1}{2} = 0$$

Therefore, $|+\rangle, |-\rangle$ are orthogonal

Normality:

$$|+\rangle = \alpha|0\rangle + \beta|1\rangle \text{ where } \alpha = \beta = \frac{1}{\sqrt{2}}. \text{ It's norm is } \sqrt{\alpha^2 + \beta^2} = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{1} = 1$$

$$|-\rangle = \alpha|0\rangle + \beta|1\rangle \text{ where } \alpha = \frac{1}{\sqrt{2}}, \beta = -\frac{1}{\sqrt{2}}. \text{ It's norm is } \sqrt{\alpha^2 + \beta^2} = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{1} = 1$$

Therefore $|+\rangle, |-\rangle$ can form an orthonormal pair of basis quantum states.

- (b) Yes. For any pair of qubits $|\phi_1\rangle, |\phi_2\rangle$ to act as a basis, they must be orthonormal to each other (orthogonal to each other and have unit norm). Since $|+\rangle, |-\rangle$ satisfy the criteria, they can also act as a valid pair of basis states to represent a qubit.
- (c) We can represent qubits in terms of $|+\rangle, |-\rangle$ as follows:

$$(i) |+\rangle + |-\rangle = \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \right) + \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right) = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle + \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = \frac{2}{\sqrt{2}}|0\rangle.$$

$$\text{Therefore, } |0\rangle = \frac{\sqrt{2}}{2}(|+\rangle + |-\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$(ii) |+\rangle - |-\rangle = \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \right) - \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right) = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle - \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = \frac{2}{\sqrt{2}}|1\rangle.$$

$$\text{Therefore, } |1\rangle = \frac{\sqrt{2}}{2}(|+\rangle - |-\rangle) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

(iii) using the representations above, we can represent any arbitrary qubit as follows:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \alpha \times \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) + \beta \times \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle) = \frac{\alpha}{\sqrt{2}}|+\rangle + \frac{\alpha}{\sqrt{2}}|-\rangle + \frac{\beta}{\sqrt{2}}|+\rangle - \frac{\beta}{\sqrt{2}}|-\rangle = \frac{\alpha+\beta}{\sqrt{2}}|+\rangle + \frac{\alpha-\beta}{\sqrt{2}}|-\rangle$$

Problem 4

The Pauli Matrices are given by:

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Show that the following relations hold:

(a) (3 points) $HX = ZH$,

(b) (3 points) $HY = -YH$,

(c) (3 points) $HZ = XH$.

where H is the Hadamard operation.

(1 point) Using these properties, show that $HXHYHZ = ZYXH$

Solution

- (a) we know that $H = \frac{1}{\sqrt{2}}V$, where $V = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. As we can pull any scalar factor out of a matrix multiplication, it is sufficient to show that $VX = ZV$.

$$VX = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 \times 0 + 1 \times 1 & 1 \times 1 + 1 \times 0 \\ 1 \times 0 - 1 \times 1 & 1 \times 1 - 1 \times 0 \end{pmatrix} = \begin{pmatrix} 0 + 1 & 1 + 0 \\ 0 - 1 & 1 - 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

$$ZV = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 0 \times 1 & 1 \times 1 + 0 \times (-1) \\ 0 \times 1 - 1 \times 1 & 0 \times 1 - 1 \times (-1) \end{pmatrix} = \begin{pmatrix} 1 + 0 & 1 + 0 \\ 0 - 1 & 0 - (-1) \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

- (b) using similar reasoning as above, we only need to show that $VY = -YV$

$$VY = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 \times 0 + 1 \times i & 1 \times (-i) + 1 \times 0 \\ 1 \times 0 - 1 \times i & 1 \times (-i) - 1 \times 0 \end{pmatrix} = \begin{pmatrix} 0 + i & -i + 0 \\ 0 - i & -i - 0 \end{pmatrix} = \begin{pmatrix} i & -i \\ -i & -i \end{pmatrix}$$

$$-YV = - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = - \begin{pmatrix} 0 \times 1 - i \times 1 & 0 \times 1 - i \times (-1) \\ i \times 1 + 0 \times 1 & i \times 1 + 0 \times (-1) \end{pmatrix} = - \begin{pmatrix} 0 - i & 0 + i \\ i + 0 & i + 0 \end{pmatrix} = - \begin{pmatrix} -i & i \\ i & i \end{pmatrix} = \begin{pmatrix} i & -i \\ -i & -i \end{pmatrix}$$

- (c) by observation, we see that $H^T = H, X^T = X, Z^T = Z$ (they are all symmetric matrices). We can use the property of matrices $(UV)^T = V^T U^T$ to prove the property.

We have shown in part (a) that $ZH = HX$. Taking transpose on both sides,

$$(ZH)^T = (HX)^T. \text{ Using the property of transpose, } H^T Z^T = X^T H^T.$$

Since H, X, Z are symmetric, we get $HZ = XH$

- (d) we know that $H^2 = \mathbb{I}$, where $\mathbb{I}$ is the identity matrix.

$$HXHYHZ = ZHHYHZ \text{ using part (a)}$$

$$= ZYHZ \text{ using the definitions of } H, \mathbb{I}$$

$$= ZYXH \text{ using part (c)}$$

Problem 5

(10 points) We have seen earlier that if we are given one of the two quantum states $|+\rangle$ and $|-\rangle$ such that it is not known whether it is a $|+\rangle$ and $|-\rangle$, we can distinguish between them *perfectly* by applying a Hadamard operation followed by a measurement.

Suppose we are randomly sent one of the following two states:

$$(a) \frac{1}{2} |0\rangle + \frac{\sqrt{3}}{2} i |1\rangle, (b) \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} i |1\rangle,$$

Devise a procedure to find out which one of the two states has been sent. This procedure should perform better than a random choice with a success probability as high as possible. More specifically, given one of the two qubits, by performing your operation(s) followed by a measurement, the probability of finding out which one of the states is should be as high as possible, with the lower limit being greater than a random guess, i.e. $> 1/2$.

Solution

The Qubit $\frac{1}{2} |0\rangle + \frac{\sqrt{3}}{2} i |1\rangle$ collapses upon its basis state as per the following probabilities. $|\alpha|^2 = 1/4(|0\rangle), |\beta|^2 = 3/4(|1\rangle)$

Furthermore, the qubit $\frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} i |1\rangle$ collapses upon its basis state as per the following probabilities. $|\alpha|^2 = 3/4(|0\rangle), |\beta|^2 = 1/4(|1\rangle)$

Now given that a qubit has been received, blindly guessing if its either qubit A, or B would give us the correct answer 50percent of the time.

A better approach would be to measure the qubit upon receipt, if it collapses to $|0\rangle$ we can simply assume that the original qubit was B, and A if the measurement came out to be $|1\rangle$.

This is because, A has a 75percent chance to collapse to 1 and B has a 75percent chance to collapse to 0, Therefore if we guess after the measurement, we are likely to guess correctly 75percent of the time.